

**Team Identifier - G**  
**Project Name - Coder V**

**Project Option 2- A Code Reading Educational Tool**

**Milestone 6**

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### ***What it does?***

CoderV is a web-based Python learning platform. Learners work through structured lessons and hands-on activities: bug-fix labs, visualization of code execution, data structures and algorithms content, lightweight web customization labs (HTML/CSS/JS), quizzes, and progress tracking. The app ties lessons, practice, and assessment together behind sign-in, with user data and content served from a backend API.

### ***Languages and technologies***

The frontend is a React single-page app built with Vite, styled with Tailwind CSS, and routed with React Router. Firebase handles authentication; the client talks to a Node.js Express REST API. The backend verifies Firebase ID tokens, exposes endpoints for users, progress, lessons, quizzes, and related features, and uses Supabase (PostgreSQL) for persistent data. Python is used on the server side for a dynamic code tracer (step-through execution of learner-submitted code). OpenAI is integrated via the official SDK for in-app assistance. Supporting pieces include lucide-react for icons and standard Node tooling (e.g. ESLint, tests).

### ***How it is deployed***

The frontend is hosted on Vercel, with deployments tied to the main branch. The Express backend runs on Render, deployed from its backend repository. Firebase provides hosted auth; Supabase hosts the database. OpenAI is used as a managed API from the backend for the features that call the model.

### ***Use of AI***

AI appears in the product through OpenAI: the platform uses an LLM (e.g. gpt-4o-mini in configuration) for quiz hints and a general “Ask AI” assistant so learners get contextual help without leaving the flow. We have used tools like Cursor, to make the design tweaks, like having the architecture and 80% of UI already in place, to cover all the small things like not visible in certain screen sizes, and certain colors, using AI for that.

Additional Note - We are still using the AI to see what can be the future as during the showcase Theresa Pardo(**Associate Vice President for Research & Economic Development**) showed an interest to implement for middle school, we are just thinking, but using AI to see how we can scale and make it useful.